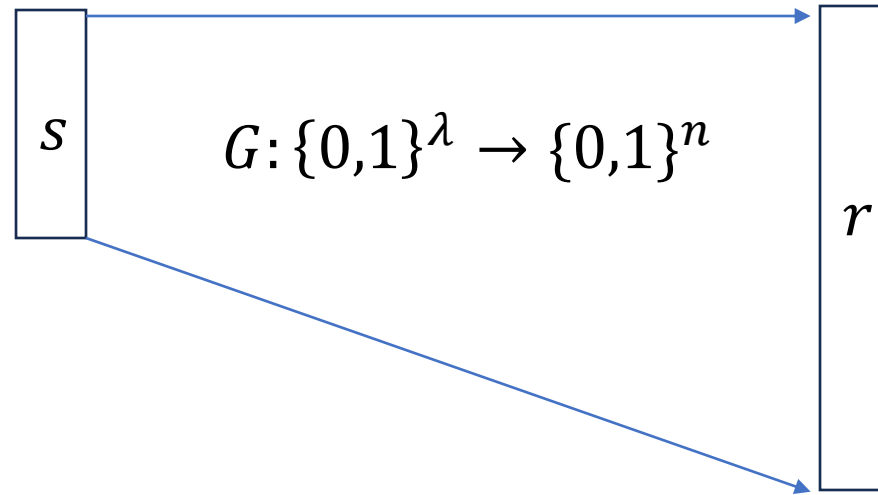


COS 433: Cryptography

Reminder:

- Problem Set 1 is out! Submit by 9/24 on Gradescope.
- If you haven't yet, submit survey + (optional) problem set 0.

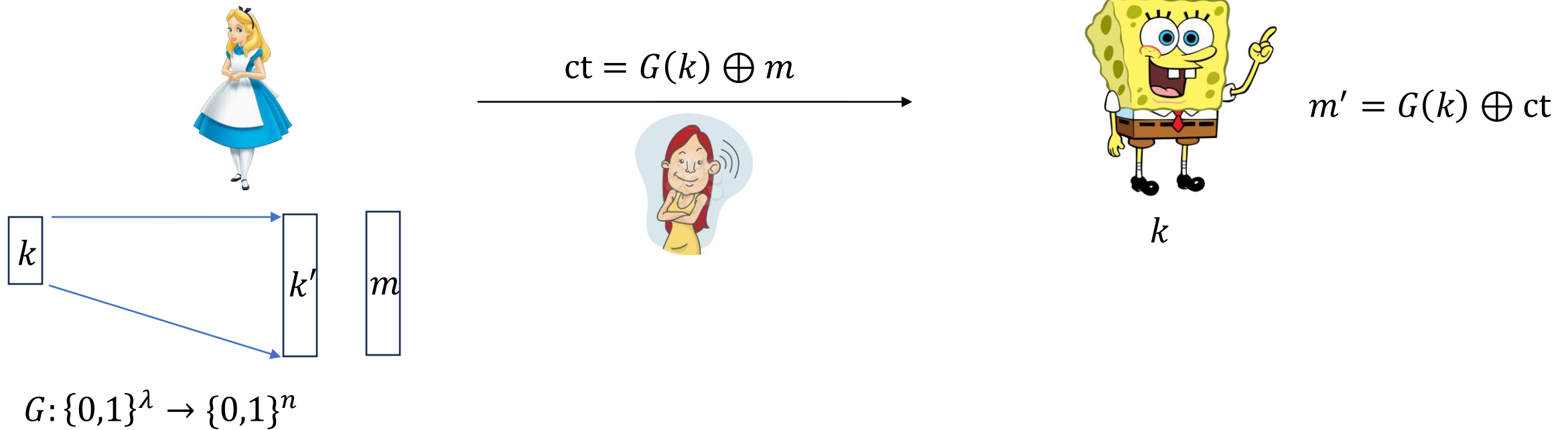
Pseudorandom Generators



Definition: An **efficiently computable** function family $G = \{G^{(\lambda)}: \{0,1\}^\lambda \rightarrow \{0,1\}^{n(\lambda)}\}_\lambda$ is a **pseudorandom generator (PRG)** if it satisfies two additional properties:

- Stretch: $n(\lambda) > \lambda$ for all λ
- Pseudorandomness: $G(U_\lambda) \approx_c U_n$ (uniform n bit string)

A construction!



Theorem: if G is a PRG, then this is a secure one-time encryption scheme!

Security “Proof by Reduction”



k, m

$$ct = G(k) \oplus m$$



k

$$m' = G(k) \oplus ct$$

Proof: Want to show that if G is a PRG, then $\text{Enc}(k, m_0) \approx_c \text{Enc}(k, m_1)$

Hybrid argument

$$G(\textcolor{red}{k}) \oplus \textcolor{blue}{m}_0 \approx_c U_n \approx_c G(\textcolor{red}{k}) \oplus \textcolor{blue}{m}_1$$

Security “Proof by Reduction”

Want to show that if G is a PRG, then $\text{Enc}(k, m_0) \approx_c \text{Enc}(k, m_1)$

Claim: for all message tuples $m = (m^{(\lambda)})_\lambda$, $G(k) \oplus m \approx_c U_n$

Proof: suppose D is an efficient distinguisher violating the claim:

$$|\mathbf{E}[D(G(k) \oplus m)] - \mathbf{E}[D(r)]| = \epsilon(\lambda)$$

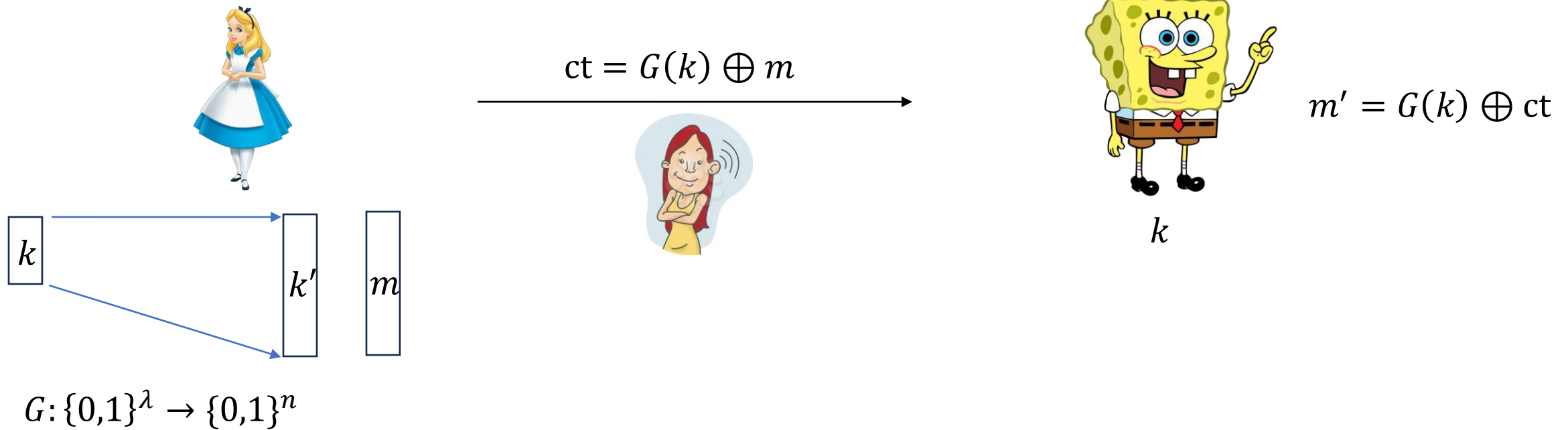
Then, we claim that $G(k)$ can be efficiently distinguished from U_n .

$$D'(x) = D(x \oplus m)$$

$$|\mathbf{E}[D'(G(k))] - \mathbf{E}[D'(r)]| = \epsilon(\lambda)$$

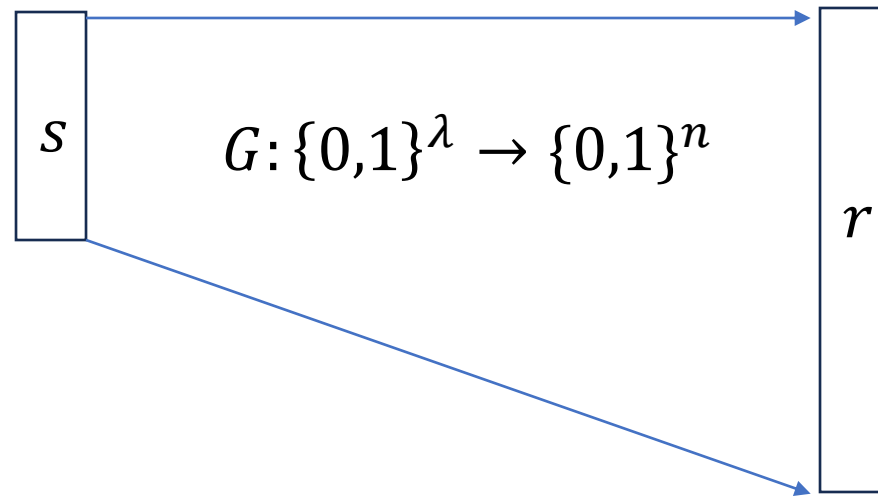
contradicts the security of G

A construction!



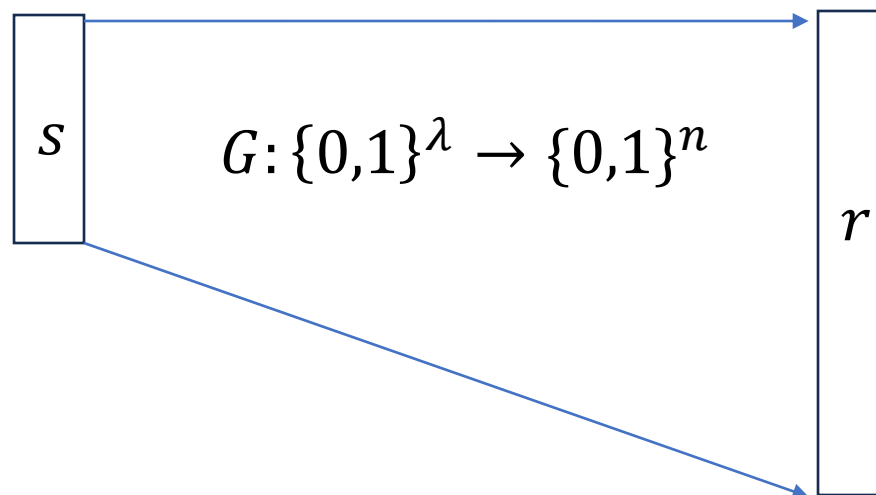
Corollary: if PRGs exist, then non-trivial encryption schemes exist!

Pseudorandom Generators



Do PRGs exist?

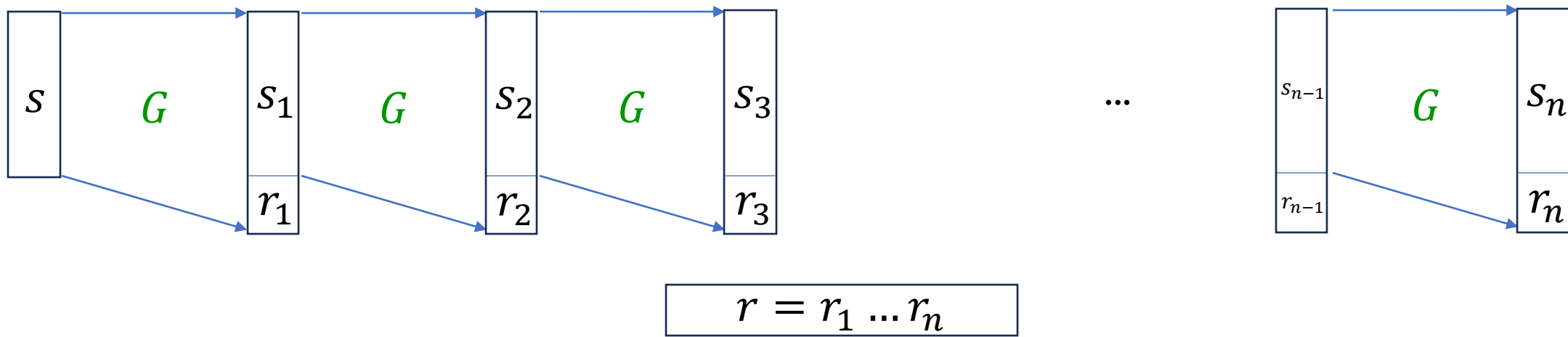
Pseudorandom Generators



Theorem: 1 bit of “stretch” is enough!

If $G: \{0,1\}^\lambda \rightarrow \{0,1\}^{\lambda+1}$ is a PRG, then for any $n(\lambda) = \lambda^c$, there exists $G': \{0,1\}^\lambda \rightarrow \{0,1\}^{n(\lambda)}$

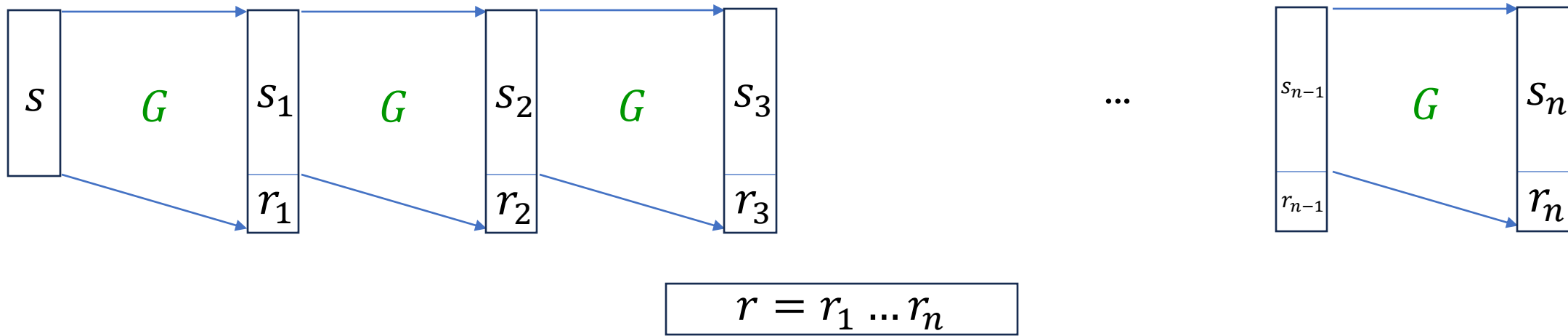
PRG Length Extension



Theorem: 1 bit of “stretch” is enough!

If $G: \{0,1\}^\lambda \rightarrow \{0,1\}^{\lambda+1}$ is a PRG, then for any $n(\lambda) = \lambda^c$, there exists $G': \{0,1\}^\lambda \rightarrow \{0,1\}^{n(\lambda)}$

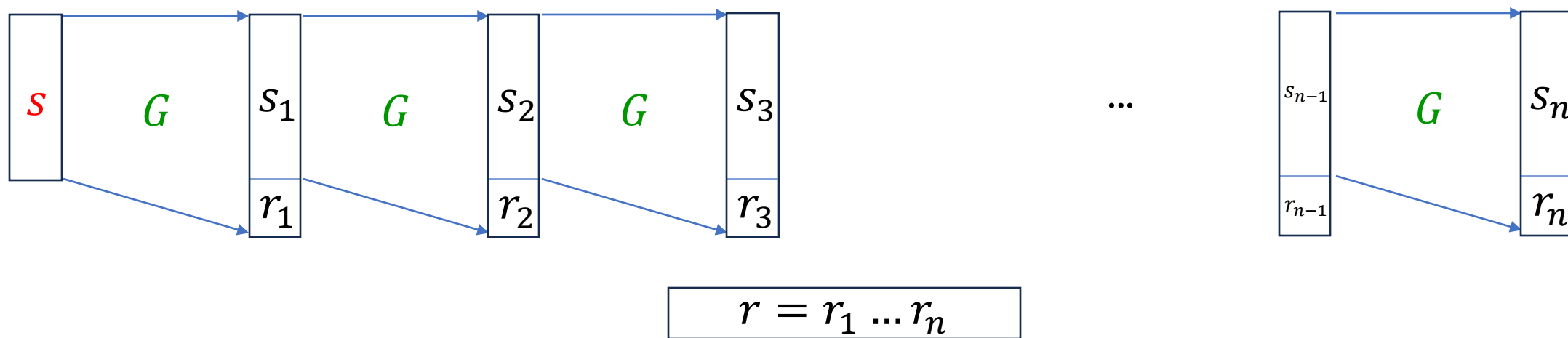
PRG Length Extension



Claim: $G'(s) \approx_c r$

Proof: Another proof by reduction

PRG Length Extension

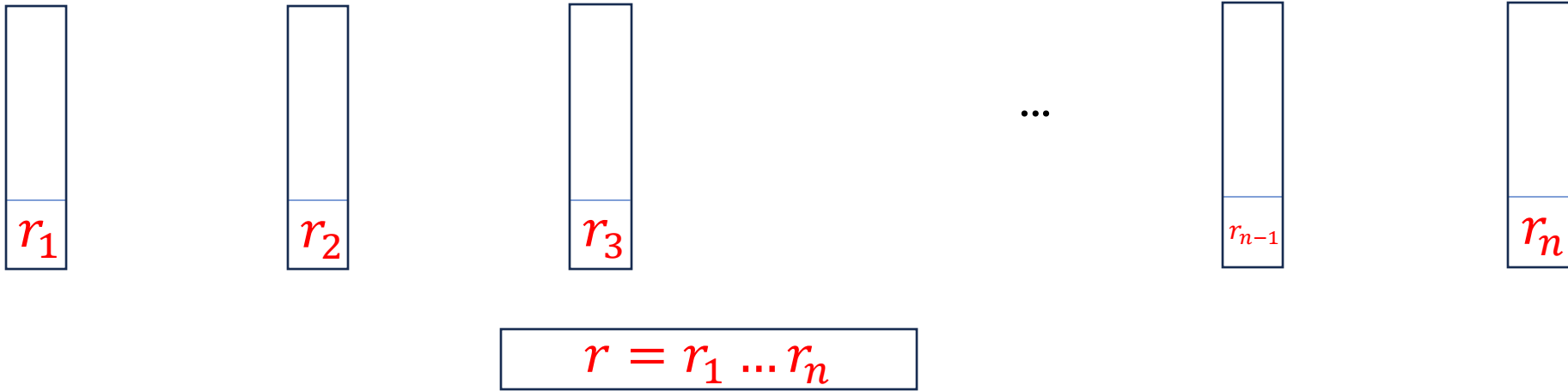


Claim: $G'(s) \approx_c r$

Proof: Define the following distributions

$$X_0^{(\lambda)} = G'(U_\lambda)$$

PRG Length Extension

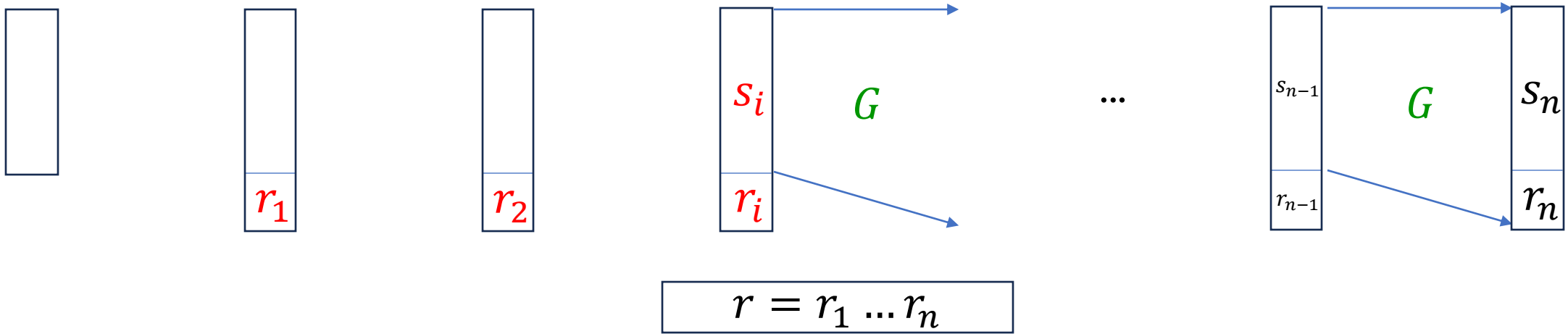


Claim: $G'(s) \approx_c r$

Proof: Define the following distributions

$$X_n^{(\lambda)} = U_n$$

PRG Length Extension

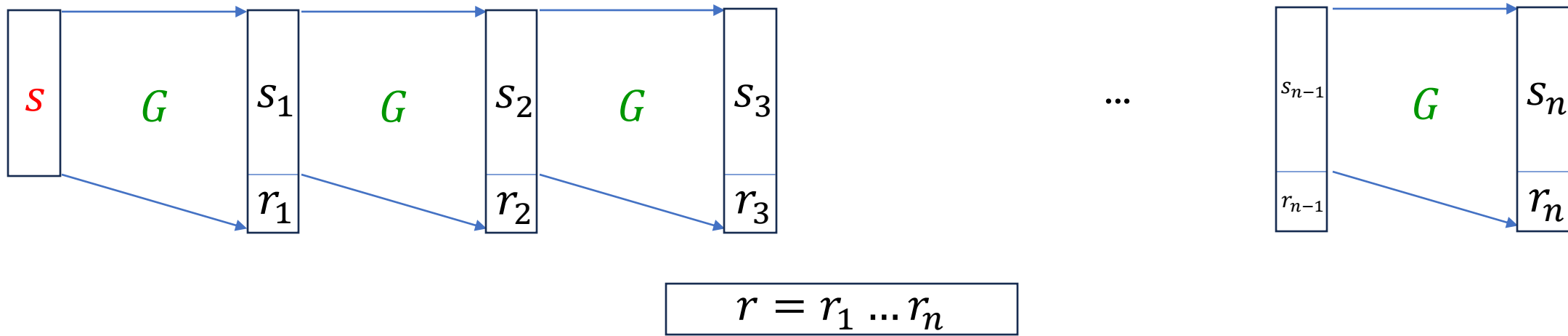


Claim: $G'(s) \approx_c r$

Proof: Define the following distributions

$$X_i^{(\lambda)} = U_i \times G'_i(U_\lambda)$$

PRG Length Extension

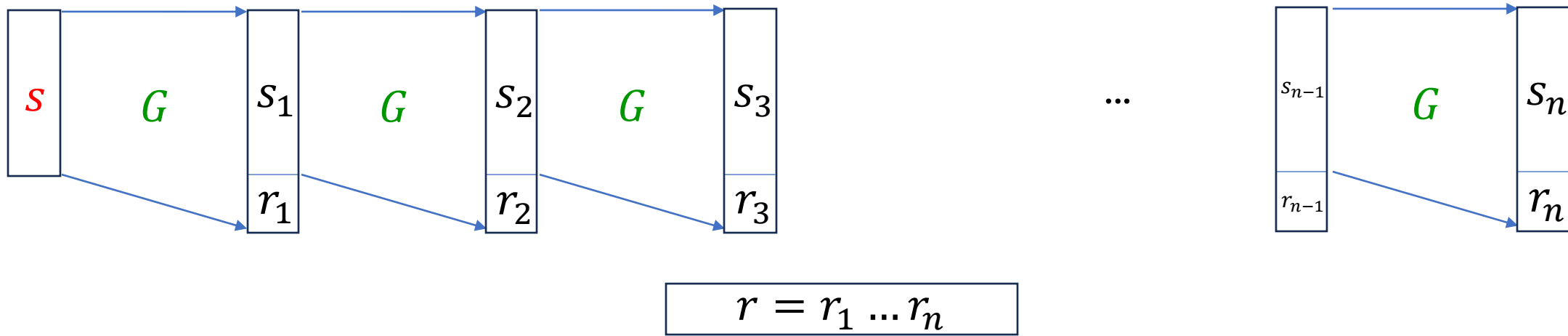


Claim: $G'(s) \approx_c r$

Proof: Define the distributions $X_i^{(\lambda)} = U_i \times G'_i(U_\lambda)$. Then,

$$|\mathbf{E}[D(G'(s))] - \mathbf{E}[D(r)]| \leq \sum_{i=1}^n |\mathbf{E}[D(X_i^{(\lambda)})] - \mathbf{E}[D(X_{i-1}^{(\lambda)})]|$$

PRG Length Extension

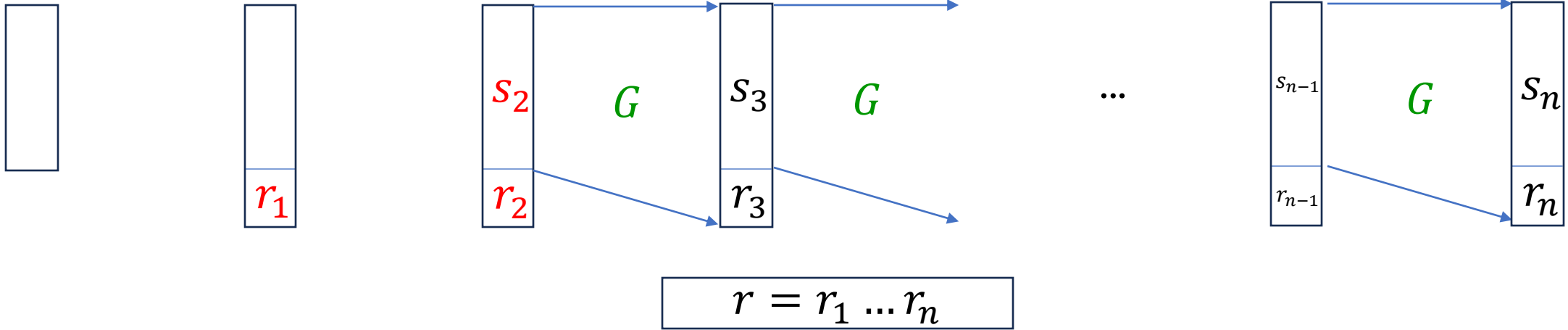


Claim: $G'(s) \approx_c r$

Proof: Define the distributions $X_i^{(\lambda)} = U_i \times G'_i(U_\lambda)$. Then, $\exists i$ such that

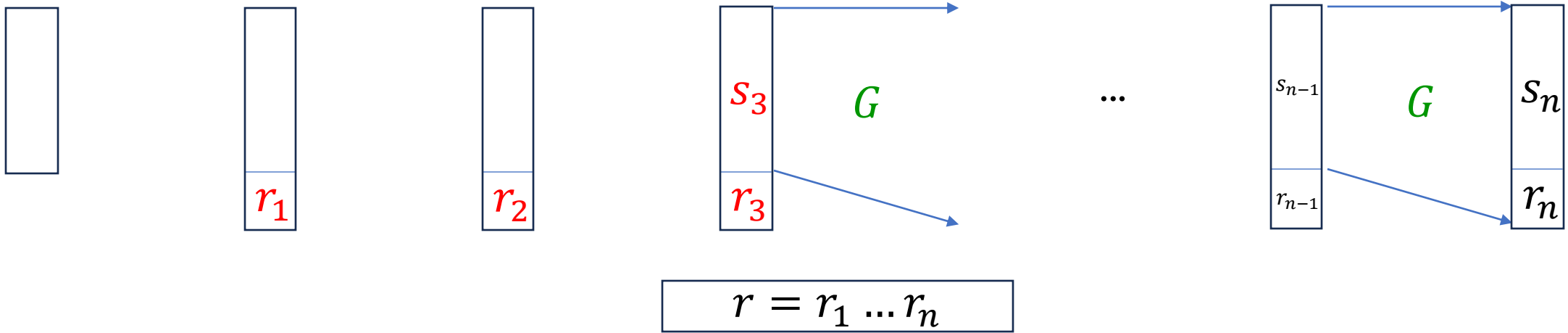
$$\left| \mathbf{E} \left[D \left(X_i^{(\lambda)} \right) \right] - \mathbf{E} \left[D \left(X_{i-1}^{(\lambda)} \right) \right] \right| \geq \frac{1}{n(\lambda)} \cdot \left| \mathbf{E} [D(G'(s))] - \mathbf{E} [D(r)] \right|$$

PRG Length Extension



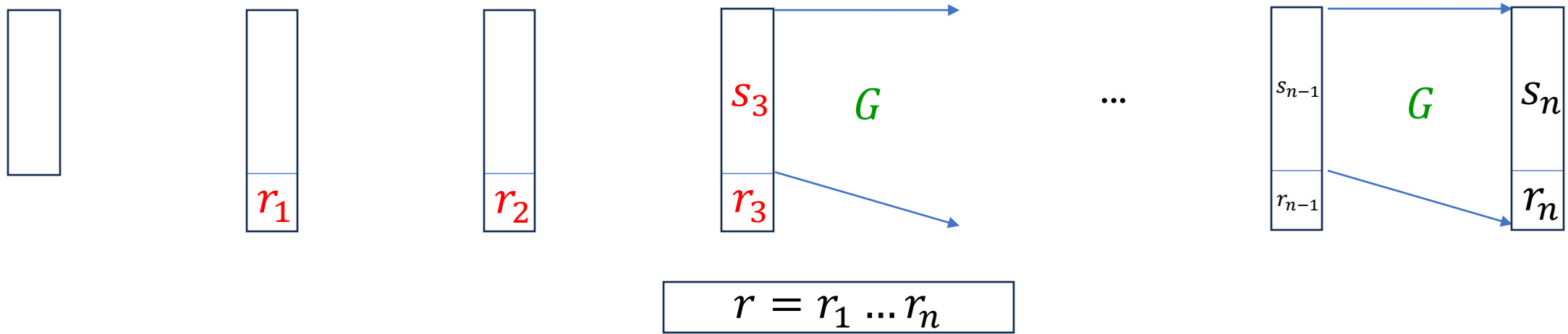
$$X_{i-1}^{(\lambda)} = U_{i-1} \times G'_{i-1}(U_\lambda)$$

PRG Length Extension



$$X_i^{(\lambda)} = U_i \times G'_i(U_\lambda)$$

PRG Length Extension



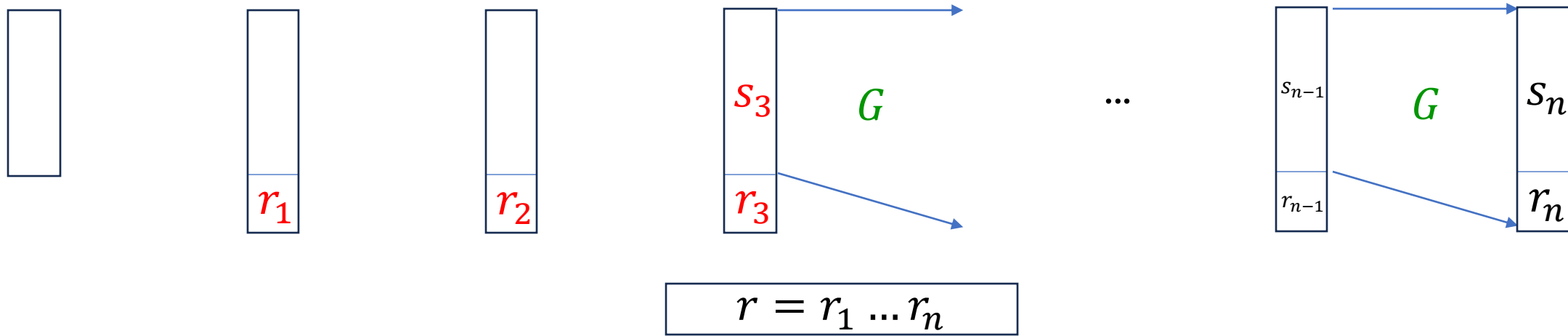
Proof: If G' is insecure, we will show that G is broken.

Define:

$$D_i^{(\lambda)}(x) = D(\textcolor{red}{r}_1, \dots \textcolor{red}{r}_{i-1}, x_{\lambda+1}, \textcolor{blue}{G}'_i(\textcolor{blue}{x}_{1:\lambda}))$$

$$\left| \mathbf{E} \left[D_i^{(\lambda)}(\textcolor{red}{r}_i, \textcolor{red}{s}_i) \right] - \mathbf{E} \left[D_i^{(\lambda)}(\textcolor{green}{G}(\textcolor{red}{s}_{i-1})) \right] \right| = \left| \mathbf{E} \left[D \left(\textcolor{blue}{X}_i^{(\lambda)} \right) \right] - \mathbf{E} \left[D \left(\textcolor{blue}{X}_{i-1}^{(\lambda)} \right) \right] \right|$$

PRG Length Extension



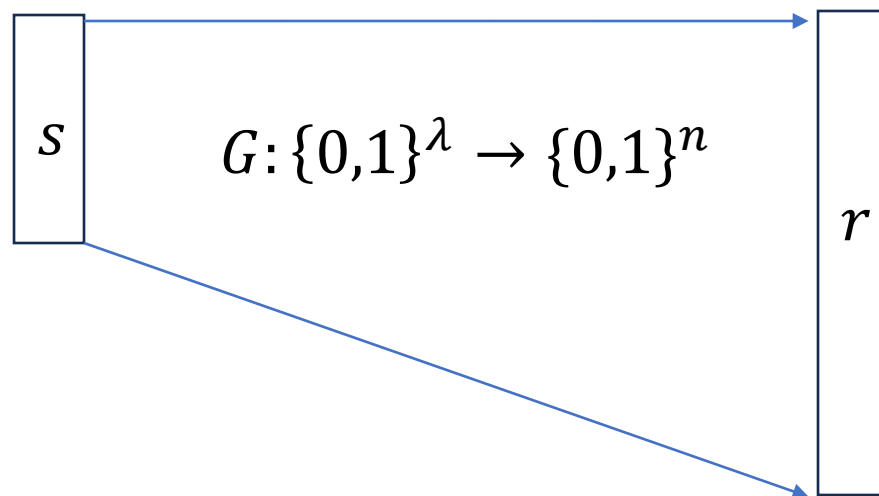
Proof: If G' is insecure, we will show that G is broken.

Define:

$$D_i^{(\lambda)}(x) = D(r_1, \dots, r_{i-1}, x_{\lambda+1}, G'_i(x_{1:\lambda}))$$

$$\left| \mathbf{E} \left[D_i^{(\lambda)}(r_i, s_i) \right] - \mathbf{E} \left[D_i^{(\lambda)}(G(s_{i-1})) \right] \right| \geq \frac{1}{n(\lambda)} \cdot |\mathbf{E}[D(G'(s))] - \mathbf{E}[D(r)]|$$

Pseudorandom Generators

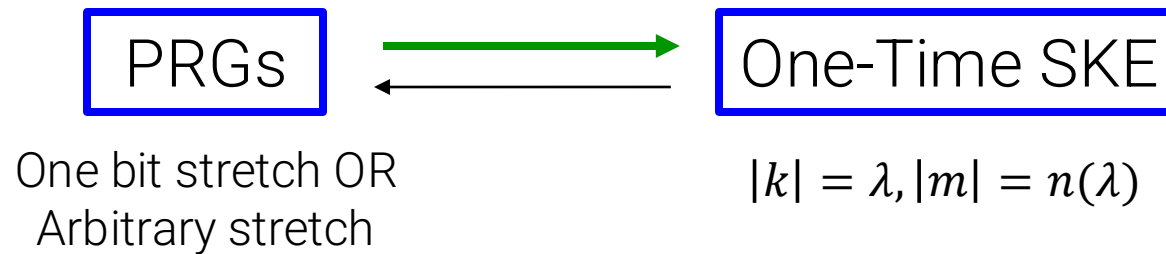


Theorem: 1 bit of “stretch” is enough!

If $G: \{0,1\}^\lambda \rightarrow \{0,1\}^{\lambda+1}$ is a PRG, then for any $n(\lambda) = \lambda^c$, there exists $G': \{0,1\}^\lambda \rightarrow \{0,1\}^{n(\lambda)}$

Relationships between Cryptographic Primitives

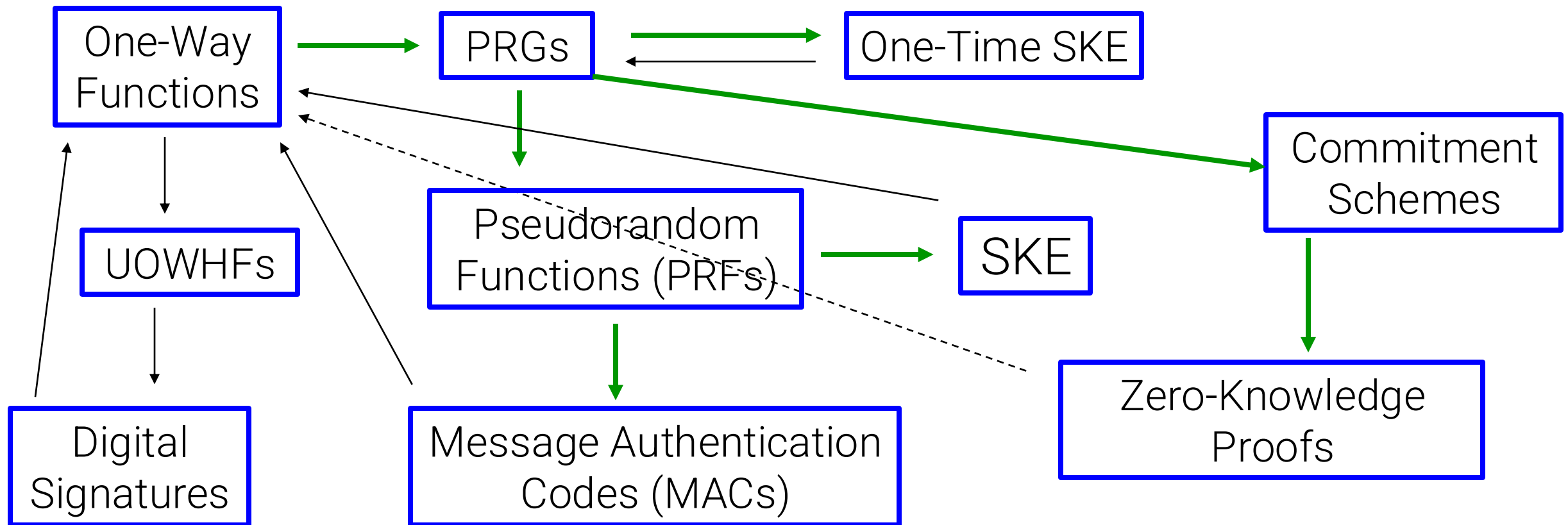
Theorem: the following are **existentially equivalent** (we won't prove this)



To build one-time SKE (for long messages), it is **necessary and sufficient** to construct a PRG!

Relationships between Cryptographic Primitives

Theorem: the following are **existentially equivalent** (we won't prove this in full)



“The PRG Conjecture”

Main Conjecture: PRGs with one bit of stretch exist

If true, we have secret-key cryptography!

“The PRG Conjecture”

Main Conjecture: PRGs with one bit of stretch exist

Evidence for conjecture:

- 1) **Equivalent** to many other conjectures (including weaker looking ones)
- 2) “**Generic attack**” that treats G as a black box shouldn’t be expected to work (HW1)
- 3) **Concrete candidates!** Have withstood attacks for many decades (or centuries...)
 - from practice (AES)
 - from theory (next slide)

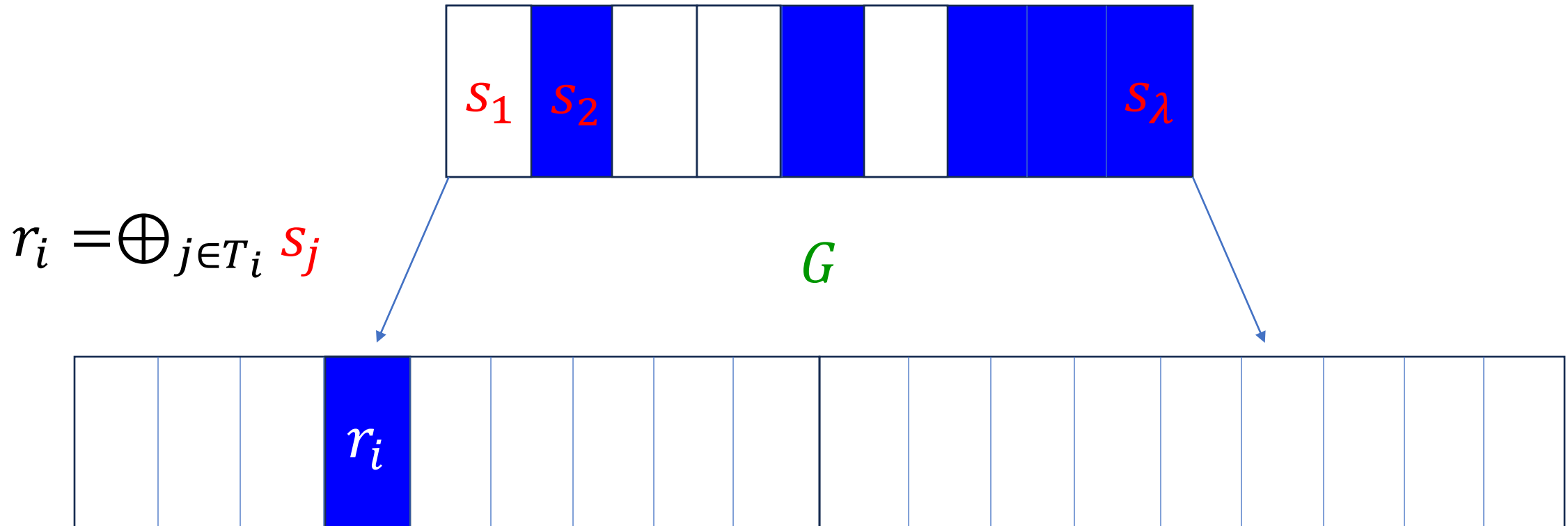
Let's build a PRG!

The Subset Sum Generator

s_1	s_2							s_λ
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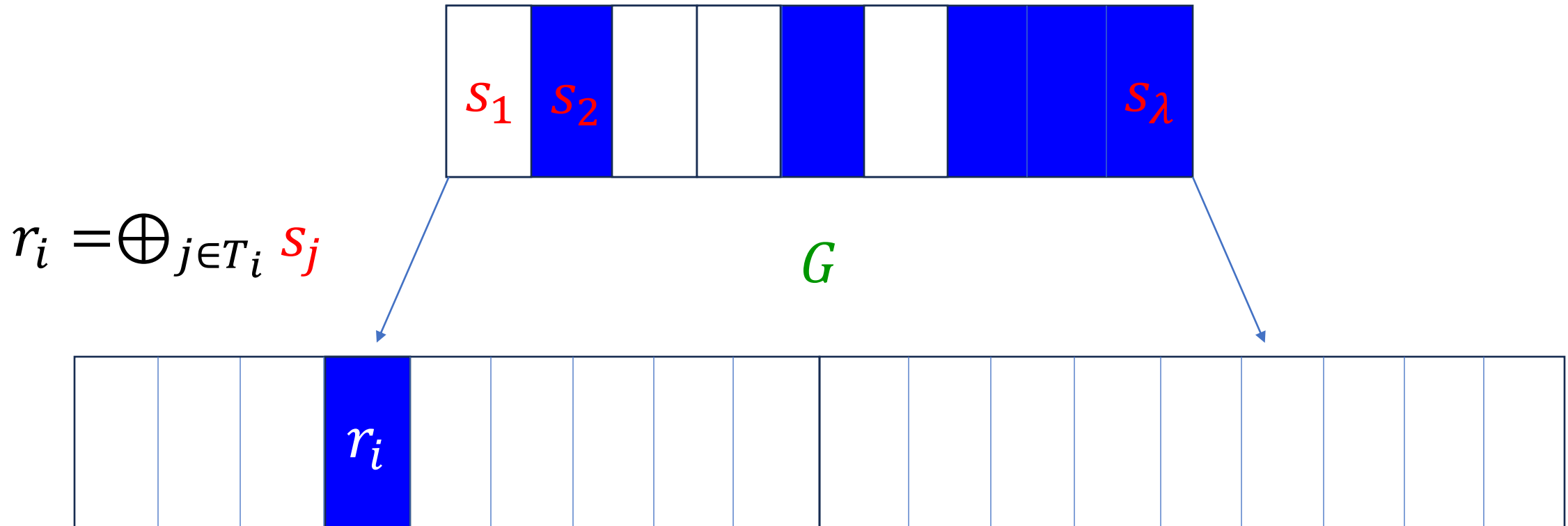
The Subset Sum Generator

A subset sum generator is described by a collection of sets $T_1, \dots, \mathbf{T}_i, \dots, T_{n(\lambda)} \subset [\lambda]$



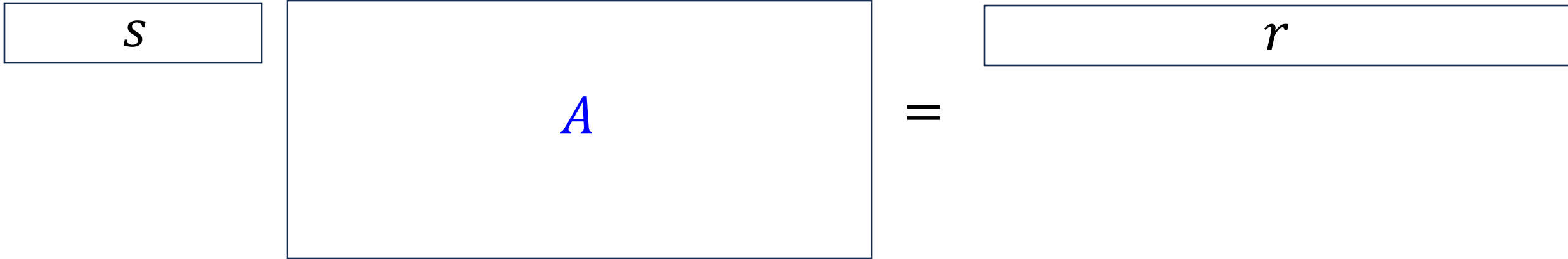
The Subset Sum Generator

Very commonly used in practice... but not for cryptography!



The Subset Sum Generator

$$G_A(\boxed{s}) = sA \pmod{2}$$



The Subset Sum Generator

$$G_A(\boxed{s}) = sA \pmod{2}$$

Poly-time attack:

- Given $y \in \{0,1\}^n$, check if $sA \equiv y \pmod{2}$ has a solution (“column reduction”)
- In the pseudorandom case, there will always be a solution!
- But for a uniformly random r , $sA \equiv r \pmod{2}$ is solvable w.p. at most $2^{\lambda-n}$

Takeaway: need some non-linearity in your PRG

To be continued